



Swipe back closed apps

With Android 11, you will be able to bring back closed apps by swiping down from the top of the screen. <https://digit.in/may20-29>



Auto-complete dev code

Codata has picked up a nice \$12M for an AI platform that auto-completes developers' code. <https://digit.in/may20-30>

pursue creating some form of a solid-state device. He decided to challenge a number of colleagues such as William Shockley, Jack Scaff, Russell Ohl and others to investigate semiconductor technology to replace the vacuum tubes. He knew that the vacuum tubes were far too unreliable to be the ultimate answer to seamless electronics.

Russel Ohl began looking into silicon crystals and how the preparation of silicon crystals could influence



Source: IEEE Journal of Solid-State Circuits

John Bardeen, William Shockley and Walter Brattain at Bell Labs

if you get an N or P-type silicon. The direction in which the rectification occurred defined the type of the silicon. Ohl was also able to develop a sample where the top part was a p-type region while the bottom was n-type. He found that when the light was shone onto this, it developed a voltage.

In 1945 after the World War II, Shockley began experimenting with improved semiconductor materials that were developed for radar detectors used in the war. He investigated a field-effect amplifier, one that was similar to concepts patented by Austro-Hungarian physicist, Julius E. Lilienfeld in 1926 and German physicist, Dr Oskar Heil in 1934. The concepts created by these two physicists failed to work as intended in the past. However, the breakthrough came when physicist John Bardeen suggested that the electrons present on the surface of the semiconductor could be obstructing the penetration of electric fields into the material. Along with William Shockley's support and the assistance

of another physicist, Walter Brattain, John Bardeen began researching the behaviour of these 'surface states'.

By late November 1947, Bardeen and Brattain managed to create a working transistor, a crude one which they continued to improve until December 16, 1947. On this day, their research finally culminated in creating a working 'point-contact' transistor made out of two gold foil contacts sitting on a germanium crystal. It was also known as the 'PNP point-contact germanium transistor'. They made a circuit to demonstrate the working solid-state device to Bell Labs on December 23. Bardeen and Brattain also spent some time filing patents for their invention and clearing it for public release with the military. In June 1948, Bell Labs officially announced the revolutionary solid-state device which they called a 'transistor'.

Shockley, however, went on to building a more reliable and reproducible version of the first working transistor created by Bardeen and Brattain. He was successful and introduced his bipolar junction transistor in 1952. It was made from a solid piece of semiconductor material with no point contacts. This invention was superior to the point-contact type and replaced it by dominating the industry for the next 30 years. He applied for a patent for the transistor effect and a transistor amplifier as well. In 1956, all three Bell Labs physicists, William Shockley, Walter Brattain and John Bardeen were awarded the Nobel Prize in Physics for their contributions.



Source: Wikipedia

A replica of the first transistor invented at Bell Labs in 1947

IMPROVEMENTS AND COMMERCIAL ADOPTION

The junction transistor was first used in a commercial product, a Sonotone hearing aid, in 1952. Following this, in 1954, the very first transistor radio, the Regency TR1 was manufactured. Different manufacturing methods were constantly being developed to produce faster, cheaper and more reliable transistors. Another important advancement was the silicon transistor made in 1954, first by Morris Tanenbaum at Bell Labs and shortly after that, by a team at Texas Instruments, led by chemist Willis Adcock. By the end of the 50s, silicon was the preferred material in the industry for manufacturing transistors and Texas Instruments became the dominant semiconductor vendor.

A start-up in Silicon Valley, California, created the Fairchild Semiconductor. In 1958, they introduced a double-diffused silicon mesa transistor which became a huge commercial success. Shortly after, Swiss physicist Jean Hoerni created a revolutionary planar process which transformed the manufacturing process of transistors from a semi-handcrafted operation to a high-volume automated production process. This also paved the way for the development of the modern ICs (integrated circuits).

In 1959, the ideas of Lilienfeld and Heil inspired a Korean electrical engineer, Dawon Kahng, to build the first successful field-effect transistor (FET) which consisted of layers of metal, oxide and silicon. The MOS-FET (Metal, Oxide, Silicon field-effect transistor), often shortened to MOS, promised a much smaller, cheaper and lower power transistor. Companies such as Fairchild and RCA introduced commercial MOS transistors to the public in 1964. However, merely a decade later they were largely replaced by ICs in computers. MOS transistors, however, proved to be the most practical approach when it came to building high-density ICs. Now, billions of transistors that are manufactured every day are MOS devices. 4